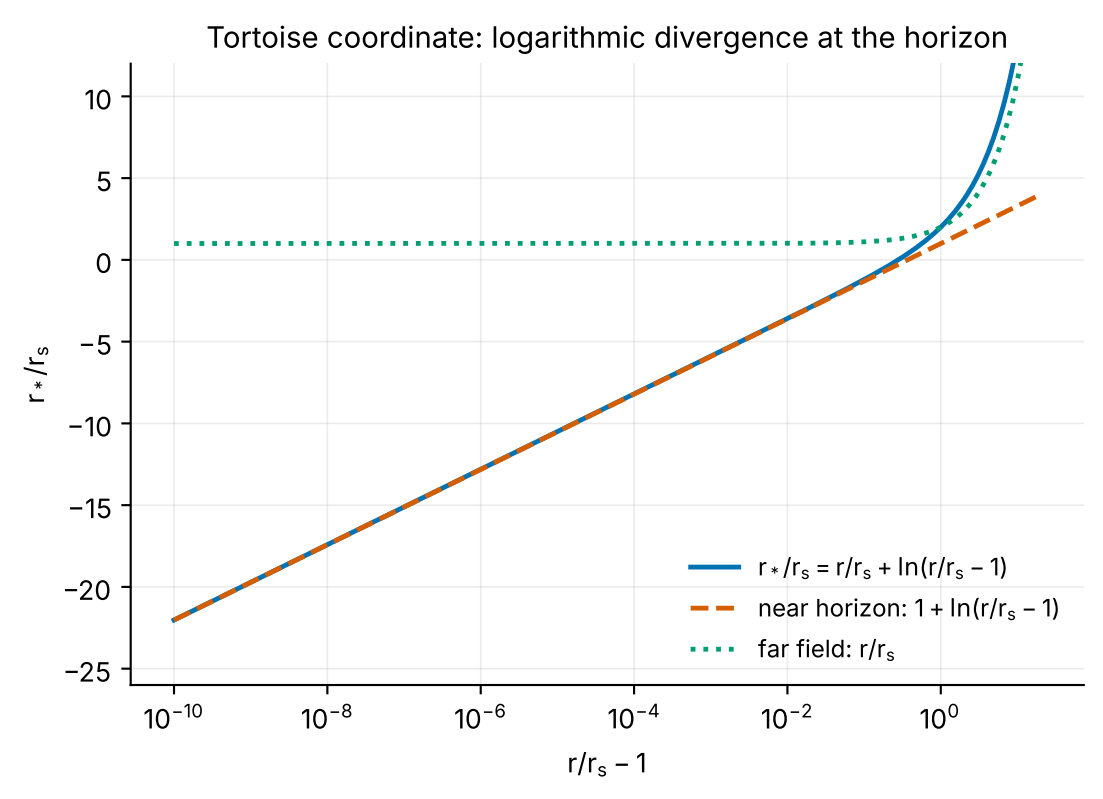
Tortoise coordinate: logarithmic divergence at the horizon
10
5
0
−5
−10
−15
−20
−25
r∗/rs
10−10
10−8
10−6
10−4
10−2
100
r/rs − 1
r∗/rs = r/rs + ln(r/rs − 1)
near horizon: 1 + ln(r/rs − 1)
far field: r/rs